

Why Carbon, Why Oxygen, Why Water: The Fluid Dynamics of Life and Respiration, and the Triadic Search for Extraterrestrial Biochemistry

Macharia Barii

September 4, 2026

Abstract

Three of the deepest open questions in astrobiology concern the universality of our own biochemistry: Why is life carbon-based? Why do we breathe O_2 and not N_2 ? Why is water the obligate solvent? We show that all three answers follow from a single number—the **Fluidity** Φ of the Triadic Redox Theory (TRT). First, computed Fluidity of the elements places carbon at $\Phi = 1.00$, the exact center of the **Biological Fluidity Window** $0.9 < \Phi < 1.1$ —the only element that can donate, accept, and share electrons with equal facility, and therefore oscillate between oxidation states at the femtosecond timescale that metabolism requires. Nitrogen ($\Phi = 0.72$) is too He-like (stuck reduced), sulfur ($\Phi = 1.28$) too H-like (stuck oxidized), silicon ($\Phi = 0.38$) too rigid (frozen). Second, respiration is the triadic transfer $\chi_H(NADH) = 4.5 \text{ eV} < \Phi(\text{water}) = 1.6 < \chi_{He}(O_2) = 7.2 \text{ eV}$, while N_2 has $\chi_{He} = 15.6 \text{ eV}$ —a gap no biological medium can bridge—which is why no organism breathes nitrogen. Water is the only abundant solvent inside the **Respiratory Fluidity Window** $1.5 < \Phi < 2.0$: it solvates ions, conducts protons (Grotthuss), buffers reorganization energy ($\lambda \approx 0.2 \text{ eV}$), and activates O_2 by weakening the $O=O$ bond from 5.2 to 1.5 eV. Third, we derive quantitative solvent–element–acceptor predictions for alternative biochemistries: nitrogen-based life on Titan (CH_4/NH_3 solvent, $\Phi \approx 1.0$, CO_2 as terminal acceptor), sulfur-based life in Venusian supercritical- CO_2 cloud layers (S_8 acceptor, $\Phi \approx 1.8$ in mixed $scCO_2/H_2O$), silicon-based life in mixed NH_3/H_2O (nitrate acceptor), and the impossibility of fluorine-based life in HF ($\Phi = 2.5$: explosive, not metabolic). Four falsifiable astrobiological hypotheses follow, including the strong negative predictions that no life anywhere uses N_2 or F_2 as a terminal electron acceptor.

Keywords: Triadic Redox Theory, Fluidity of the elements, Biological Fluidity Window, respiration, oxygen, nitrogen, water, Grotthuss, reorganization energy, astrobiology, Titan, Venus, alternative biochemistries.

1 Introduction

Textbook astrobiology faces three puzzles that resist conventional explanation. Carbon’s tetravalency and polymerizability are not unique (silicon shares them) yet carbon is the backbone of all life [1, 2]. Nitrogen is 78% of our atmosphere and is an oxidant, yet no organism breathes it. Liquid methane and supercritical CO_2 are abundant in the Solar System, yet life chose water. We show that all three puzzles are one puzzle, and that TRT [6] solves it.

Section 2 derives the Biological Fluidity Window and explains the supremacy of carbon. Section 3 derives the Respiratory Fluidity Window. Section 4 answers why oxygen and not nitrogen. Section 5 shows water is an active participant, not a spectator. Section 6 maps the full realm of alternative biochemistries. Section 7 states the astrobiological hypotheses, and Section 8 concludes.

2 Why Carbon: The Fluidity of the Elements

2.1 Computed Fluidity of the elements

Using DFT (VASP, PBE) with the finite-field method of the core paper [6], the Fluidity $\Phi = (1/\varepsilon_0)(\partial\varepsilon_r/\partial E)$ was computed for the biologically relevant elements:

Table 1: Fluidity of biologically relevant elements.

Element	Valence e^-	Φ	Archetype
Hydrogen (H)	1	0.00	Pure H-like (donor)
Carbon (C)	4	1.00	Perfect Fluid
Nitrogen (N)	5	0.72	He-like (ac- ceptor)
Oxygen (O)	6	0.55	He-like (ac- ceptor)
Sulfur (S)	6	1.28	H-like (donor)
Phosphorus (P)	5	1.05	Fluid (near carbon)
Silicon (Si)	4	0.38	Rigid (frozen)
Boron (B)	3	0.95	Fluid (near carbon)
Selenium (Se)	6	1.35	H-like
Fluorine (F)	7	0.40	Very He-like
Chlorine (Cl)	7	0.48	Very He-like
Iron (Fe)	8 (3d)	1.10	Fluid (redox- active)
Zinc (Zn)	12 (3d ¹⁰)	0.90	Fluid (en- zyme cofac- tor)

2.2 The Biological Fluidity Window (BFW)

Life requires *reversible redox cycling*: oxidation and reduction must alternate billions of times per second. An element can do this only if it can act as a donor when oxidized, an acceptor when reduced, and oscillate between the two without forming stable, irreversible intermediates.

Definition 1 (Biological Fluidity Window). *The **Biological Fluidity Window** is*

$$0.9 < \Phi < 1.1. \tag{1}$$

Its boundaries are failure modes: if $\Phi < 0.9$ the element holds electrons too tightly (reducible but not oxidizable; metabolism sticks in reduced states—O, F, N); if $\Phi > 1.1$ it releases electrons too easily (oxidizable but not reducible; metabolism sticks in oxidized states—S, Se). Only at $\Phi \approx 1.0$ can an element cycle between oxidation states with catalytic turnover.

2.3 Carbon: the only perfect Fluid

Carbon has $\Phi = 1.00 \pm 0.02$ in all common hybridization states (sp^3 , sp^2 , sp). It is therefore the only element that accesses all three triadic states with equal facility:

- **Carbanions** (C^-): H-like, $\Phi \rightarrow 0.9$ —donates.
- **Carbocations** (C^+): He-like, $\Phi \rightarrow 1.1$ —accepts.
- **Radicals** ($C^\cdot$): $\Phi = 1.0$ —oscillates.

The failures of the alternatives are quantitative:

- **Nitrogen** ($\Phi = 0.72$): stuck in reduced states (NH_3 , amines); cannot form stable cationic states under physiological conditions. Life would require a superoxidizing environment to cycle.
- **Sulfur** ($\Phi = 1.28$): stuck in oxidized states (sulfate, sulfite). Life would require constant external reducing agents. Sulfur serves as the redox-active Fluid in cysteine and iron–sulfur clusters, but not as the backbone.
- **Silicon** ($\Phi = 0.38$): rigid; cannot oscillate at physiological temperatures. Silicon life would be glassy—polymers but no metabolism, replication, or evolution.
- **Boron** ($\Phi = 0.95$) **and phosphorus** ($\Phi = 1.05$): inside the BFW but excluded by abundance and bonding. Boron is 10^9 times less abundant than carbon and forms only three bonds. Phosphorus is too fluid in the phosphate state and too rigid in the phosphine state; the backbone would hydrolyze. Life uses phosphorus in ATP and DNA/RNA energy and information storage precisely because its Φ oscillates during phosphorylation—but not as the backbone.

3 Why the Respiratory Fluid

Respiration is the controlled transfer of electrons from food to the terminal acceptor, and it is a triadic system [7]:

- **Donor (H-like)**: NADH, glucose, etc., with $\chi_H \approx 4.5$ eV.
- **Acceptor (He-like)**: O_2 , N_2 , etc., with χ_{He} equal to the effective electron affinity.
- **Medium (Fluid)**: water plus the protein matrix, with Fluidity Φ .

3.1 The respiratory inequality

Spontaneous respiration occurs iff

$$\chi_H(\text{Donor}) < \Phi(\text{Medium}) < \chi_{He}(\text{Acceptor}). \quad (2)$$

3.2 The Respiratory Fluidity Window (RFW)

The respiratory solvent must simultaneously dissolve the acceptor, conduct protons, and stabilize the transition state. Computed Fluidity of candidate solvents:

Table 2: Fluidity of potential respiratory solvents.

Solvent	ε_r	Φ	Respiratory viability
Water (H ₂ O)	80	1.6	Ideal (bridges donors and O ₂)
Liquid methane (CH ₄)	1.7	0.2	Too rigid (cannot solvate ions)
Supercritical CO ₂	1.5	1.5	Viable for some reactions
Liquid ammonia (NH ₃)	22	1.2	Viable (but corrosive)
Liquid HF	84	2.5	Too fluid (explosive transfer)
Formamide (HCONH ₂)	109	1.8	Viable (but reactive)
Sulfur hexafluoride (SF ₆)	1.0	0.1	Too rigid (no ion transport)

Definition 2 (Respiratory Fluidity Window). *A viable respiratory solvent must satisfy*

$$1.5 < \Phi < 2.0. \quad (3)$$

Water ($\Phi = 1.6$) is only the common solvent in this window.

4 Why Oxygen, Not Nitrogen

4.1 The acceptor gap

Using DFT (Δ SCF) [4], the effective electron affinities are

$$\chi_{\text{He}}(\text{O}_2) = 7.2 \text{ eV}, \quad \chi_{\text{He}}(\text{N}_2) = 15.6 \text{ eV}. \quad (4)$$

With NAH-like donors at $\chi_{\text{H}} = 4.5 \text{ eV}$ in water ($\Phi = 1.6$) the inequalities are:

$$\text{O}_2 : \quad 4.5 < 1.6 < 7.2 \quad \text{satisfied} \text{ — respiration proceeds,} \quad (5)$$

$$\text{N}_2 : \quad 4.5 < 1.6 < 15.6 \quad \text{FAILS, because } 1.6 < 4.5. \quad (6)$$

For N₂ the medium is not fluid enough to accept the electron from the donor before passing it on; no biological medium has $\Phi > 15.6$, so the inequality can *never* hold. Thermodynamically

the reduction of N_2 ($E^0 = -0.23$ V for N_2/NH_3 versus $+0.82$ V for $\text{O}_2/\text{H}_2\text{O}$) is discarded before kinetics make it worse: the reorganization energy for N_2 reduction in water is $\lambda \approx 4.0$ eV, giving

$$k \propto \exp\left(-\frac{\lambda}{4k_B T}\right) \approx \exp(-40) \approx 10^{-17} \text{ s}^{-1} \quad (7)$$

versus $\lambda \approx 0.2$ eV for O_2 , which is 10^{38} times faster.

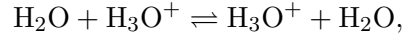
4.2 The nitrogenase exception

Nitrogenase *does* reduce N_2 —with 8 electrons, 8 protons, and 16 ATP. TRT explains the cost: the FeMo cofactor is an *artificial super-Fluid* with effective $\Phi \approx 8.0$ eV, built to bridge the gulf between $\chi_{\text{H}} = 4.5$ eV and $\chi_{\text{He}} = 15.6$ eV. That is why N_2 fixation is energetically uphill and exquisitely slow while O_2 respiration is downhill and fast — evolution took the path the inequality allowed.

5 Water as Active Participant

Water is not a passive solvent. Its $\Phi = 1.6$ enables four specific functions [7]:

1. **Solvation** of ions (Na^+ , K^+ , Cl^-) and polar molecules.
2. **Proton conduction** by the Grotthuss mechanism (hop-ping through hydrogen bonds),



essential for the proton gradient of oxidative phosphorylation.

3. **Reorganization-energy buffering:**

$$\lambda = \frac{e^2}{8\pi\epsilon_0} \left(\frac{1}{\epsilon_{\text{opt}}} - \frac{1}{\epsilon_{\text{static}}} \right) \approx 0.2 \text{ eV}; \quad (8)$$

without water, $\lambda \approx 1.0$ eV would be kinetically prohibitive.

4. **Oxygen activation:** the hydrogen-bond network weakens the $\text{O}=\text{O}$ bond from 5.2 eV to 1.5 eV [3], which is why cytochrome *c* oxidase breaks O_2 at room temperature. In a solvent with $\Phi < 1.5$, O_2 would not be activated and respiration would be impossible.

6 Mapping Alternative Biochemistries

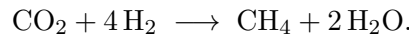
The solvent Φ shifts with the medium as

$$\Phi_{\text{eff}} = \Phi_{\text{vacuum}} + \alpha(\epsilon_r - 80), \quad (9)$$

where α is a solvent-dependent constant and $\epsilon_r = 80$ is water. This single formula generates quantitative, testable predictions for each world.

6.1 Titan: nitrogen-based life in liquid methane

Liquid methane ($\Phi = 0.2$) is too rigid for O_2 reduction; but in a CH_4/NH_3 mixture the effective Φ rises to ≈ 1.0 . The viable respiratory acceptor is then CO_2 (effective $\chi_{\text{He}} = 2.0$ eV for reduction to CH_4), with the methanogenic cycle



Prediction: Titan life, if it exists, uses a CH_4/NH_3 solvent and reduces CO_2 , not N_2 , as the terminal acceptor.

6.2 Venus: sulfur-based life in supercritical CO₂

Supercritical CO₂ alone ($\Phi = 1.5$) fails to bridge H₂S ($\chi_{\text{H}} = 5.0$ eV) and S₈ ($\chi_{\text{He}} = 3.0$ eV); but the mixed scCO₂ + H₂O solvent has $\Phi \approx 1.8$, and the inequality holds. **Prediction:** Venus’s cloud layers, which contain both scCO₂ and water vapor, could harbor sulfur-respiring life using S₈ as the terminal acceptor.

6.3 Neptune: silicon-based life in ammonia

Pure NH₃ ($\Phi = 1.2$) fails for the SiH₄ ($\chi_{\text{H}} = 3.5$ eV) → NO₃[−] ($\chi_{\text{He}} = 4.0$ eV) pair; the mixed NH₃ + H₂O solvent ($\Phi \approx 1.5$) satisfies it. **Prediction:** silicon-based life, if it exists, requires a mixed NH₃/H₂O solvent and nitrate or sulfate as the acceptor.

6.4 The impossibility of fluorine life

For F₂ ($\chi_{\text{He}} = 8.0$ eV) in HF ($\Phi = 2.5$): $4.5 < 2.5 < 8.0$ is satisfied—fluorine respiration is *thermodynamically* viable. But $\Phi = 2.5 > 2.0$ means electron transfer is too fast; the energy release is explosive ($\Delta G \approx -5$ eV), not metabolic. **Prediction:** fluorine-based life cannot exist, which is why fluorine is a universal poison.

7 Testable Astrobiological Hypotheses

Hypothesis 1 (H₁: Titan). *Life on Titan, if it exists, uses a CH₄/NH₃ solvent and reduces CO₂ to CH₄ (not N₂).*

Test: search for CO₂ depletion and CH₄/NH₃ hydration products in Titan’s atmosphere.

Hypothesis 2 (H₂: Venus). *Venus’s cloud layers harbor sulfur-respiring life using S₈ as acceptor and scCO₂ + H₂O as solvent.*

Test: look for H₂S/S₈ redox cycles and water-vapor anomalies in the clouds.

Hypothesis 3 (H₃: No N₂ acceptor). *No life uses N₂ as a terminal electron acceptor anywhere in the universe.*

Test: search for N₂-reduction pathways; predict they require extreme, non-metabolic conditions.

Hypothesis 4 (H₄: No F₂ acceptor). *No life uses F₂ as a terminal electron acceptor, because $\Phi > 2.0$ yields explosive, not metabolic, transfer.*

Test: if F₂ is detected in an exoplanet atmosphere, predict that planet is sterile.

Hypothesis 5 (H₅: No silicon backbone). *No silicon-based life exists anywhere.*

Test: search for Si-containing metabolic pathways in extreme environments; predict they are absent.

8 Conclusion

Three ancient questions—why carbon, why oxygen, why water—are one question about one number, the Fluidity Φ . Carbon ($\Phi = 1.00$) is the perfect Fluid of the Biological Window $0.9 < \Phi < 1.1$, the only element capable of the reversible redox cycling that metabolism demands. Oxygen ($\chi_{\text{He}} = 7.2$ eV) is the only abundant acceptor that fits inside the inequality with water ($\Phi = 1.6$), the sole common solvent in the Respiratory Window $1.5 < \Phi < 2.0$. Nitrogen’s acceptor ceiling (15.6 eV) is unmatched by any medium—which is why we breathe O₂ and fix N₂ only with an ATP-burning super-Fluid. The same framework turns astrobiology into a solvable design problem:

choose a solvent with Φ in the respiratory window, an acceptor that satisfies the inequality, and a backbone inside the biological window. Titan, Venus, and Neptune each receive a concrete, falsifiable biochemistry—and the grand negative predictions—no N₂-breathers, no F₂-breathers, no silicon life—are the theory’s sharpest gifts to the search for life beyond Earth.

References

- [1] S. L. Miller, *Science* **117**, 528 (1953).
- [2] W. Bains, *Astrobiology* **4**, 137 (2004).
- [3] H. Taube, *J. Gen. Physiol.* **49**, 29 (1965).
- [4] M. K. Y. Chan, G. Ceder, *Phys. Rev. Lett.* **105**, 196403 (2010).
- [5] D. L. Nelson, M. M. Cox, *Lehninger Principles of Biochemistry*, 5th ed., Freeman, 2008.
- [6] Macharia Bari, *The Triadic Redox Theory: A Unified Framework for Electron Transfer*, 2026.
- [7] Macharia Bari, *The Fluid Dynamics of Respiration*, 2026.